

ADVANCED VECTOR CALCULUS

Vector and Scalar Point Function

1. Find the velocity and acceleration of moving particle along the curve $\vec{r} = (t, t^2, t^3)$ at $t=1$.

Here,

$$\vec{r} = t\vec{i} + t^2\vec{j} + t^3\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = \vec{i} + 2t\vec{j} + 3t^2\vec{k}$$

$$\text{and } \frac{d^2\vec{r}}{dt^2} = 2\vec{j} + 6t\vec{k}$$

At $t=1$,

$$\text{velocity} = \frac{d\vec{r}}{dt} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\text{acceleration} = \frac{d^2\vec{r}}{dt^2} = 2\vec{j} + 6\vec{k} \quad \checkmark$$

2. Find the angle between the curve $\vec{r} = (t^2, 3t^2, -(t^2+1))$ at the point $t = \pm 1$.

Here,

$$\vec{r} = t^2\vec{i} + 3t^2\vec{j} - (t^2+1)\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = 2t\vec{i} + 6t\vec{j} - 2t\vec{k}$$

At $t=1$,

$$T_1 = 2\vec{i} + 6\vec{j} - 2\vec{k}$$

At $t=-1$,

$$T_2 = -2\vec{i} - 6\vec{j} + 2\vec{k}$$

Let θ be the angle between T_1 and T_2 , then,

$$\cos \theta = \frac{\vec{T}_1 \cdot \vec{T}_2}{|\vec{T}_1| |\vec{T}_2|} = \frac{-4 - 36 - 4}{\sqrt{4+36+4} \sqrt{4+36+4}} = -1$$

$$\therefore \theta = \cos^{-1}(-1)$$

$$= \underline{\underline{\pi}}$$

3. A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t$. Find the component of velocity and acceleration at $t=1$ along $\vec{i} + \vec{j} + 3\vec{k}$.

sol given curve is

$$\vec{r} = (t^3 + 1)\vec{i} + t^2\vec{j} + (2t + 5)\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = 3t^2\vec{i} + 2t\vec{j} + 2\vec{k}$$

and

$$\frac{d^2\vec{r}}{dt^2} = 6t\vec{i} + 2\vec{j} + 0$$

At $t=1$,

$$\frac{d\vec{r}}{dt} = 3\vec{i} + 2\vec{j} + 2\vec{k}$$

$$\frac{d^2\vec{r}}{dt^2} = 6\vec{i} + 2\vec{j}$$

Also, given vector is

$$\vec{a} = \vec{i} + \vec{j} + 3\vec{k}$$

$$\therefore \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{i} + \vec{j} + 3\vec{k}}{\sqrt{1+1+9}} = \frac{1}{\sqrt{11}} (\vec{i} + \vec{j} + 3\vec{k})$$

Then,

The component of velocity at $t=1$ along $\hat{a}$ is

$$\frac{d\vec{r}}{dt} \cdot \hat{a} = (3\vec{i} + 2\vec{j} + 2\vec{k}) \cdot \frac{1}{\sqrt{11}} (\vec{i} + \vec{j} + 3\vec{k})$$

at

$$= \frac{3+2+6}{\sqrt{11}} = \frac{11}{\sqrt{11}} = \underline{\underline{\sqrt{11}}}$$

The component of acceleration at $t=1$ along $\hat{a}$ is

$$\frac{d^2 \vec{r}}{dt^2} \cdot \hat{a} = (6\vec{i} + 2\vec{j}) \cdot \frac{1}{\sqrt{11}} (\vec{i} + \vec{j} + 3\vec{k})$$

$$= \frac{6+2}{\sqrt{11}} = \underline{\underline{8/\sqrt{11}}}$$

✓

4. If $\vec{r} = (3t, 3t^2, 2t^3)$ then show that $[\vec{r} \ \ddot{\vec{r}} \ \dddot{\vec{r}}] = 216$.

Here,

$$\vec{r} = 3t\vec{i} + 3t^2\vec{j} + 2t^3\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = \dot{\vec{r}} = 3\vec{i} + 6t\vec{j} + 6t^2\vec{k}$$

$$\frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}} = 0\vec{i} + 6\vec{j} + 12t\vec{k}$$

$$\frac{d^3\vec{r}}{dt^3} = \dddot{\vec{r}} = 0\vec{i} + 0\vec{j} + 12\vec{k}$$

Then,

$$[\vec{r} \ \ddot{\vec{r}} \ \dddot{\vec{r}}] = \begin{vmatrix} 3 & 6t & 6t^2 \\ 0 & 6 & 12t \\ 0 & 0 & 12 \end{vmatrix}$$

$$= 3(72 - 0) - 6t(0 - 0) + 6t^2(0 - 0)$$

$$= 216 \quad \underline{\text{proved}}$$

5. Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at $(1, 2, 3)$

along $\vec{a} = \vec{i} + \vec{j} + \vec{k}$

given surface is

$$f = x^2 - y^2 + 2z^2$$

$$\therefore \nabla f = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (x^2 - y^2 + 2z^2)$$

$$= 2x\vec{i} - 2y\vec{j} + 4z\vec{k}$$

At $(1, 2, 3)$,

$$\nabla f = 2\vec{i} - 4\vec{j} + 12\vec{k}$$

Also, given vector is

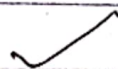
$$\vec{a} = \vec{i} + \vec{j} + \vec{k}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{1+1+1}} = \frac{1}{\sqrt{3}} (\vec{i} + \vec{j} + \vec{k})$$

Then,

The directional derivative of f at $(1, 2, 3)$ along $\vec{a}$ is

$$\begin{aligned} D_{\vec{a}}(f) &= \nabla f \cdot \hat{a} \\ &= (2\vec{i} - 4\vec{j} + 12\vec{k}) \cdot \frac{1}{\sqrt{3}} (\vec{i} + \vec{j} + \vec{k}) \\ &= \frac{2 - 4 + 12}{\sqrt{3}} \\ &= \frac{10}{\sqrt{3}} \end{aligned}$$



6. Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at $P(1, 2, 3)$ along PQ where Q is the point $(5, 0, 4)$.

Given surface is

$$f = x^2 - y^2 + 2z^2$$

$$\begin{aligned} \therefore \nabla f &= \left(\vec{i} \frac{d}{dx} + \vec{j} \frac{d}{dy} + \vec{k} \frac{d}{dz} \right) (x^2 - y^2 + 2z^2) \\ &= 2x\vec{i} - 2y\vec{j} + 4z\vec{k} \end{aligned}$$

At $P(1, 2, 3)$,

$$\nabla f = 2\vec{i} - 4\vec{j} + 12\vec{k}$$

Also, we have,

$$\begin{aligned}\vec{PQ} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (5 - 1, 0 - 2, 4 - 3) \\ &= (4, -2, 1) \\ &= 4\vec{i} - 2\vec{j} + \vec{k}\end{aligned}$$

$$\hat{PQ} = \frac{\vec{PQ}}{|\vec{PQ}|} = \frac{4\vec{i} - 2\vec{j} + \vec{k}}{\sqrt{16 + 4 + 1}} = \frac{1}{\sqrt{21}} (4\vec{i} - 2\vec{j} + \vec{k})$$

Then,

The directional derivative of f at $(1, 2, 3)$ along $\hat{PQ}$ is

$$\begin{aligned}D_{\hat{PQ}}(f) &= \nabla f \cdot \hat{PQ} \\ &= (2\vec{i} - 4\vec{j} + 12\vec{k}) \cdot \frac{1}{\sqrt{21}} (4\vec{i} - 2\vec{j} + \vec{k}) \\ &= \frac{8 + 8 + 12}{\sqrt{21}} \\ &= \frac{28}{\sqrt{21}}\end{aligned}$$

7. Find the directional derivative of $f = 4xz^3 - 3x^2yz^2$ at $(2, -1, 2)$ along z -axis (i.e. $\vec{a} = \vec{k}$).

Given surface is

$$f = 4xz^3 - 3x^2yz^2$$

$$\nabla f = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (4xz^3 - 3x^2yz^2)$$

$$= (4z^3 - 6xyz^2)\vec{i} + (-3x^2z^2)\vec{j} + (12xz^2 - 6x^2yz)\vec{k}$$

At $(2, -1, 2)$

$$\nabla f = 80\vec{i} - 48\vec{j} + 144\vec{k}$$

Also, given vector is

$$\vec{a} = \vec{k}$$

$$\therefore \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \vec{k}$$

Then,

The directional derivative of f at $(2, -1, 2)$ along $\vec{a}$ is

$$\begin{aligned} D_{\vec{a}}(f) &= \nabla f \cdot \hat{\vec{a}} \\ &= [4z^3 - 6xyz^2]\vec{i} - 3x^2z^2\vec{j} + [12xz^2 - 6x^2yz]\vec{k} \cdot \vec{k} \\ &= (80\vec{i} - 48\vec{j} + 144\vec{k}) \cdot \vec{k} \\ &= 144 \end{aligned}$$

8. If $\phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$.

Given,

$$\phi = \log(x^2 + y^2 + z^2)$$

$$\therefore \text{grad } \phi = \nabla \phi$$

$$\begin{aligned} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (\log(x^2 + y^2 + z^2)) \\ &= \frac{2x}{x^2 + y^2 + z^2} \vec{i} + \frac{2y}{x^2 + y^2 + z^2} \vec{j} + \frac{2z}{x^2 + y^2 + z^2} \vec{k} \\ &= \frac{2}{x^2 + y^2 + z^2} (x\vec{i} + y\vec{j} + z\vec{k}) \end{aligned}$$

Then,

$$\text{div}(\text{grad } \phi) = \text{div}(\nabla \phi)$$

$$= \nabla \cdot (\nabla \phi)$$

$$= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot \frac{2}{x^2 + y^2 + z^2} (x\vec{i} + y\vec{j} + z\vec{k})$$

$$= \left[\frac{-x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^2} + \frac{x^2 - y^2 + z^2}{(x^2 + y^2 + z^2)^2} + \frac{x^2 + y^2 - z^2}{(x^2 + y^2 + z^2)^2} \right] \times 2$$

$$= \frac{x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^2} \times 2$$

$$= \frac{2}{x^2 + y^2 + z^2}$$

$$= \frac{2}{x^2 + y^2 + z^2}$$

$$\text{curl}(\text{grad } \phi) = \text{curl}(\nabla \phi)$$

$$= \nabla \times \nabla \phi$$

$$= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \times \frac{2}{x^2+y^2+z^2} (x\vec{i} + y\vec{j} + z\vec{k})$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{2x}{x^2+y^2+z^2} & \frac{2y}{x^2+y^2+z^2} & \frac{2z}{x^2+y^2+z^2} \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial}{\partial y} \left(\frac{2z}{x^2+y^2+z^2} \right) - \frac{\partial}{\partial z} \left(\frac{2y}{x^2+y^2+z^2} \right) \right] -$$

$$\vec{j} \left[\frac{\partial}{\partial x} \left(\frac{2z}{x^2+y^2+z^2} \right) - \frac{\partial}{\partial z} \left(\frac{2x}{x^2+y^2+z^2} \right) \right] +$$

$$\vec{k} \left[\frac{\partial}{\partial x} \left(\frac{2y}{x^2+y^2+z^2} \right) - \frac{\partial}{\partial y} \left(\frac{2x}{x^2+y^2+z^2} \right) \right]$$

$$= \vec{i} \left[\frac{-2z \cdot 2y}{(x^2+y^2+z^2)^2} + \frac{2y \cdot 2z}{(x^2+y^2+z^2)^2} \right] - \vec{j} \left[\frac{-2z \cdot 2x}{(x^2+y^2+z^2)^2} + \right.$$

$$\left. \frac{2x \cdot 2z}{(x^2+y^2+z^2)^2} \right] + \vec{k} \left[\frac{-2y \cdot 2x}{(x^2+y^2+z^2)^2} + \frac{2x \cdot 2y}{(x^2+y^2+z^2)^2} \right]$$

$$= 0$$

9. Show that $\vec{F} = (y \sin z - \sin x)\vec{i} + (x \sin z + 2yz)\vec{j} + (xy \cos z + y^2)\vec{k}$ is irrotational (conservative) and then find the scalar potential function ϕ such that $\vec{F} = \nabla \phi$.

Given,

$$\vec{F} = (y \sin z - \sin x)\vec{i} + (x \sin z + 2yz)\vec{j} + (xy \cos z + y^2)\vec{k}$$

$$\therefore \text{curl } \vec{F} = \nabla \times \vec{F}$$

$$= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \times [(y \sin z - \sin x)\vec{i} + (x \sin z + 2yz)\vec{j} + (xy \cos z + y^2)\vec{k}]$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y \sin z - \sin x & x \sin z + 2yz & xy \cos z + y^2 \end{vmatrix}$$

$$= \vec{i} (x \cos z + 2y - x \cos z - 2y) - \vec{j} (y \cos z - y \cos z) + \vec{k} (\sin z - \sin z)$$

$$= 0$$

$\therefore \vec{F}$ is irrotational (conservative).

Again,

If ϕ is a scalar potential, then

$$d\phi = \vec{F} \cdot d\vec{r}$$

$$\therefore d\phi = [(y \sin z - \sin x) \vec{i} + (x \sin z + 2yz) \vec{j} + (xy \cos z + y^2) \vec{k}] \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k})$$

$$\therefore d\phi = (y \sin z - \sin x) dx + (x \sin z + 2yz) dy + (xy \cos z + y^2) dz$$

Integrating, we get,

$$\phi = xy \sin z + \cos x + xy \sin z + y^2 z + xy \sin z + y^2 z + C$$

$$\therefore \phi = 3xy \sin z + \cos x + 2y^2 z + C$$

10. Show that the value under the integral sign (i.e. integrand) is exact in the domain and evaluate the integral:

$$(i) \int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2) dx + 9x^2 y^2 dy + (4xz + 1) dz]$$

Comparing the given integral with $\int (F_1 dx + F_2 dy + F_3 dz)$, we get

$$F_1 = 6xy^3 + 2z^2,$$

$$F_2 = 9x^2 y^2,$$

$$F_3 = 4xz + 1$$

Now,

$$\frac{\partial F_1}{\partial y} = 18xy^2$$

$$\frac{\partial F_1}{\partial z} = 4z$$

$$\frac{\partial F_2}{\partial z} = 0$$

$\frac{\partial}{\partial y}$

$\frac{\partial}{\partial z}$

$\frac{\partial}{\partial z}$

$$\frac{\partial F_2}{\partial x} = 18xy^2$$

$$\frac{\partial F_3}{\partial x} = 4z$$

$$\frac{\partial F_3}{\partial y} = 0$$

$\frac{\partial}{\partial x}$

$\frac{\partial}{\partial x}$

$\frac{\partial}{\partial y}$

$$\Rightarrow \frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} = 18xy^2, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} = 4z, \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y} = 0$$

Thus, the given integral is exact.

Then,

$$\int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2)dx + 9x^2y^2dy + (4xz+1)dz]$$

$$= \int_{(1,0,2)}^{(-2,1,3)} d[\int F_1 dx + \int (\text{terms free from } x \text{ in } F_2) dy + \int (\text{terms free from } x \text{ and } y \text{ in } F_3) dz]$$

$$= \int_{(1,0,2)}^{(-2,1,3)} d[\int (6xy^3 + 2z^2)dx + \int 0 dy + \int 1 dz]$$

$$= \int_{(1,0,2)}^{(-2,1,3)} d(3x^2y^3 + 2z^2x + z)$$

$$= [3x^2y^3 + 2xz^2 + z]_{(1,0,2)}^{(-2,1,3)}$$

$$= 3 \times 4 \times 1 + 2 \times (-2) \times 9 + 3 - (3 \times 1 \times 0 + 2 \times 1 \times 4 + 2)$$

$$= -21 - 10$$

$$= -31$$

$$(ii) \int_{(0,0,0)}^{(4,1,2)} [3ydx + 3xdy + 2zdz]$$

comparing the given integral with $\int (F_1 dx + F_2 dy + F_3 dz)$, we get,

$$F_1 = 3y,$$

$$F_2 = 3x,$$

$$F_3 = 2z$$

Now,

$$\frac{\partial F_1}{\partial y} = 3$$

$$\frac{\partial F_1}{\partial z} = 0$$

$$\frac{\partial F_2}{\partial z} = 0$$

$$\frac{\partial F_2}{\partial x} = 3$$

$$\frac{\partial F_2}{\partial x} = 0$$

$$\frac{\partial F_3}{\partial z} = 0$$

$$\frac{\partial F_3}{\partial x} = 3$$

$$\frac{\partial F_3}{\partial x} = 0$$

$$\frac{\partial F_3}{\partial y} = 0$$

$$\frac{\partial F_1}{\partial y} = 3$$

$$\frac{\partial F_1}{\partial z} = 0$$

$$\frac{\partial F_2}{\partial z} = 0$$

$$\Rightarrow \frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} = 3, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} = 0, \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y} = 0$$

Thus, the given integral is exact

Then,

$$\int_{(0,0,0)}^{(4,1,2)} [3y dx + 3x dy + 2z dz]$$

$$= \int_{(0,0,0)}^{(4,1,2)} d[F_1 dx + (\text{terms free from } x \text{ in } F_2) dy + (\text{terms free from } x \text{ or } y \text{ in } F_3) dz]$$

$$= \int_{(0,0,0)}^{(4,1,2)} d[3y dx + 0 dy + 2z dz]$$

$$= \int_{(0,0,0)}^{(4,1,2)} d(3xy + z^2)$$

$$= [3xy + z^2]_{(0,0,0)}^{(4,1,2)}$$

$$= 3 \times 4 \times 1 + 4 - 0$$

$$= 16$$

Line Integral

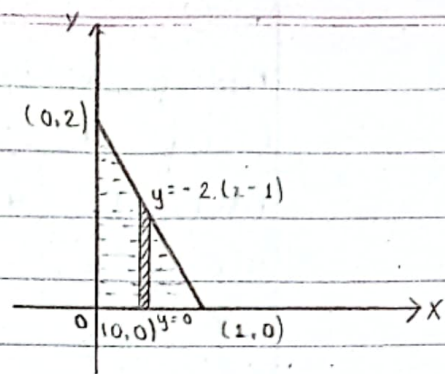
11. Evaluate $\oint_C [(x^3 - 3y) dx + (x + \sin y) dy]$, where C is the boundary of the triangle with vertices $(0,0)$, $(1,0)$, $(0,2)$.

The given integral is

$$\oint [(x^3 - 3y) dx + (x + \sin y) dy]$$

The region of integration is as shown in figure.

$$R: 0 \leq x \leq 1, \quad 0 \leq y \leq -2x+2$$



Now,

$$\oint_C [(x^3 - 3y)dx + (x + \sin y)dy]$$

$$= \iint_R \left(\frac{\partial}{\partial x} [x + \sin y] - \frac{\partial}{\partial y} [x^3 - 3y] \right) dy dx$$

$$= \int_0^1 \int_0^{2-2x} (1 + 3) dy dx$$

$$= 4 \int_0^1 [y]_0^{2-2x} dx$$

$$= 4 \int_0^1 (2-2x) dx$$

$$= 4 [2x - x^2]_0^1$$

$$= 4 (2 - 1 - 0) = 4 \times 1$$

$$= \underline{\underline{4}}$$

12 Verify Green's theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ over a region R on xy-plane bounded by lines $x=0, y=0, x=a, y=b$.

Given

$$\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$$

$$\text{Here, } F_1 = x^2 + y^2, \quad F_2 = -2xy$$

We need to verify

$$\oint_C (F_1 dx + F_2 dy) = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dy dx$$

$$\text{LHS} = \oint_C (F_1 dx + F_2 dy)$$

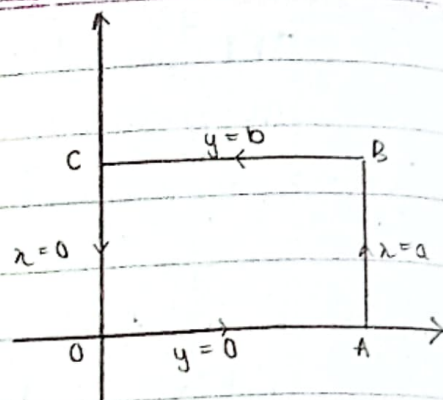
$$= \oint_C [(x^2 + y^2)dx - 2xydy]$$

Along OA, $y=0$, $dy=0$, $0 \leq x \leq a$

Along AB, $x=a$, $dx=0$, $0 \leq y \leq b$

Along BC, $y=b$, $dy=0$, $a \leq x \leq 0$

Along CO, $x=0$, $dx=0$, $b \leq y \leq 0$



Then,

$$\begin{aligned} \text{LHS} &= \int_{OA} [(x^2+y^2)dx - 2xydy] + \int_{AB} [(x^2+y^2)dx - 2xydy] + \\ &\quad \int_{BC} [(x^2+y^2)dx - 2xydy] + \int_{CO} [(x^2+y^2)dx - 2xydy] \\ &= \int_0^a (x^2+0^2)dx + \int_0^b -2aydy + \int_a^0 (x^2+b^2)dx + 0 \\ &= \left[\frac{x^3}{3} \right]_0^a - 2a \left[\frac{y^2}{2} \right]_0^b + \left[\frac{x^3}{3} + b^2x \right]_a^0 \\ &= \frac{a^3}{3} - ab^2 - \frac{a^3}{3} - ab^2 \\ &= -2ab^2 \end{aligned}$$

$$\text{RHS} = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dy dx$$

$$= \int_0^a \int_0^b \left[\frac{\partial}{\partial x} (-2xy) - \frac{\partial}{\partial y} (x^2+y^2) \right] dy dx$$

$$= \int_0^a \int_0^b (-2y - 2y) dy dx$$

$$= \int_0^a -4 \left[\frac{y^2}{2} \right]_0^b dx$$

$$= \int_0^a -2b^2 dx$$

$$= -2b^2 [x]_0^a$$

$$= -2ab^2$$

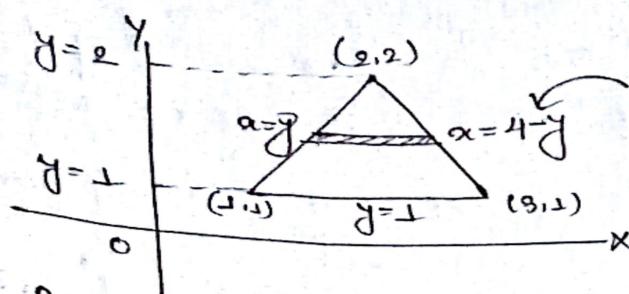
$$\therefore \text{LHS} = \text{RHS}$$

Hence, Green's theorem is verified for given $\vec{F}$.

13. Using Green's theorem, evaluate the following line integrals:

(i) $\oint_C [\sqrt{y} dx + \sqrt{x} dy]$, where C : triangle with vertices $(1,1)$, $(3,1)$ and $(2,2)$.

③ ① $\oint_C [\sqrt{y} dx + \sqrt{x} dy]$, where C : triangle $(1,1)$, $(3,1)$, $(2,2)$.



eqn of st. line passing through two points (x_1, y_1) & (x_2, y_2) is $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

$$R: y \leq x \leq 4-y$$

$$1 \leq y \leq 2$$

Now, $\oint_C [\sqrt{y} dx + \sqrt{x} dy]$

$$= \iint_R \left[\frac{\partial(\sqrt{x})}{\partial x} - \frac{\partial(\sqrt{y})}{\partial y} \right] dA$$

$$= \iint_R \left(\frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{y}} \right) dA$$

$$= \frac{1}{2} \int_1^2 \int_y^{4-y} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{y}} \right) dx dy$$

$$= \frac{1}{2} \int_1^2 \left[2\sqrt{x} - \frac{x}{\sqrt{y}} \right]_y^{4-y} dy$$

$$= \frac{1}{2} \int_1^2 \left[2\sqrt{4-y} - \frac{4-y}{\sqrt{y}} - 2\sqrt{y} + \frac{y}{\sqrt{y}} \right] dy$$

$$= \frac{1}{2} \int_1^2 \left[2\sqrt{4-y} - \frac{4}{\sqrt{y}} + \sqrt{y} - \sqrt{y} \right] dy$$

$$= \frac{1}{2} \left[2 \cdot \left(\frac{\sqrt{y}}{2} \sqrt{4-y} + \frac{4}{2} \sin^{-1} \frac{\sqrt{y}}{2} \right) - 4 \cdot 2\sqrt{y} \right]_1^2$$

$$\left[\because \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C \right]$$

$$\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$$

$$= \frac{1}{2} \left[\sqrt{y} \sqrt{4-y} + 4 \sin^{-1} \frac{\sqrt{y}}{2} - 8\sqrt{y} \right]_1$$

$$= \frac{1}{2} \left[\sqrt{2} \cdot \sqrt{2} + 4 \sin^{-1} \frac{1}{\sqrt{2}} - 8\sqrt{2} \right. \\ \left. - \left(\sqrt{3} + 4 \sin^{-1} \frac{1}{2} - 8 \right) \right]$$

$$= \frac{1}{2} \left[2 + 4 \cdot \frac{\pi}{4} - 8\sqrt{2} - \sqrt{3} - 4 \cdot \frac{\pi}{6} + 8 \right]$$

$$= \frac{1}{2} \left[-10 - 8\sqrt{2} - \sqrt{3} + \frac{\pi}{3} - \frac{2\pi}{3} \right]$$

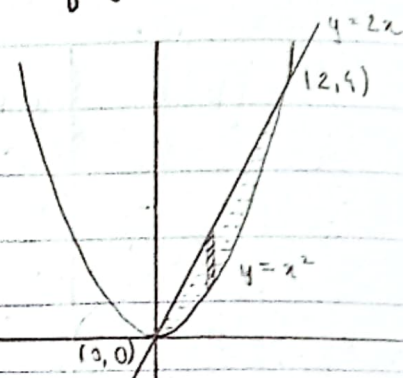
$$= \frac{1}{2} \left[-10 - 8\sqrt{2} - \sqrt{3} + \frac{\pi}{3} \right]$$

$$= 5 - 4\sqrt{2} - \frac{\sqrt{3}}{2} + \frac{\pi}{6} \quad \#$$

(ii) $\oint_C [5xy dx + x^3 dy]$, where C : region bounded by the graph $y = x^2$ and $y = 2x$ between $(0,0)$ and $(2,4)$

Here, region of integration is as shown in figure:

$$\therefore R: 0 \leq x \leq 2, x^2 \leq y \leq 2x$$



Now,

$$\oint_C [5xy dx + x^3 dy]$$

$$= \iint_R \left[\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right] dy dx \quad [\text{From Green's theorem}]$$

$$= \int_0^2 \int_{x^2}^{2x} (3x^2 - 5x) dy dx$$

$$= \int_0^2 [3x^2 y - 5xy]_{x^2}^{2x} dx$$

$$= \int_0^2 [6x^3 - 10x^2 - 3x^4 + 5x^3] dx$$

$$= \int_0^2 (-3x^4 + 11x^3 - 10x^2) dx$$

$$= \left[-3 \frac{x^5}{5} + 11 \frac{x^4}{4} - 10 \frac{x^3}{3} \right]_0^2$$

$$= -\frac{3}{5} \times 32 + \frac{11}{4} \times 16 - \frac{10}{3} \times 8 - 0$$

$$= -\frac{28}{15}$$

✓

(iii) $\oint_C [(3x^2 + y) dx + 4y^2 dy]$, where C : boundary of the triangle $(0,0)$, $(1,0)$, $(0,2)$.

Here, region of integration is as shown in figure:

$$\therefore R: 0 \leq x \leq 1, 0 \leq y \leq 2-2x$$

Now,

$$\oint_C [(3x^2 + y) dx + 4y^2 dy]$$

$$= \iint_R [\partial F_2 / \partial x - \partial F_1 / \partial y] dy dx \quad [\text{From Green's theorem}]$$

$$= \int_0^1 \int_0^{2-2x} (0 - 1) dy dx$$

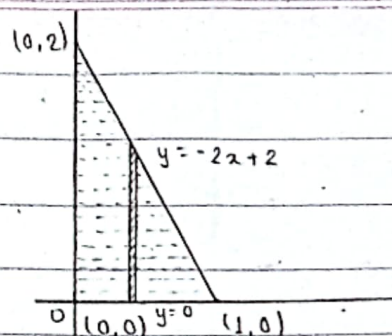
$$= - \int_0^1 [y]_0^{2-2x} dx$$

$$= - \int_0^1 (2 - 2x) dx$$

$$= - [2x - x^2]_0^1$$

$$= - (2 - 1 - 0)$$

$$= -1$$

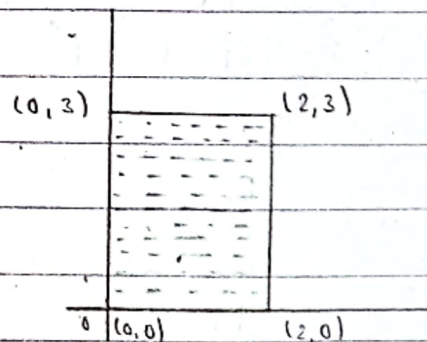


(iv) $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 e^y, y^2 e^x)$ and C : rectangle with vertices $(0,0), (2,0), (2,3), (0,3)$.

Here, the region of integration is as shown in figure.
 $\therefore R: 0 \leq x \leq 2, 0 \leq y \leq 3$

Also,

$$\begin{aligned} \vec{F} \cdot d\vec{r} &= (x^2 e^y \vec{i} + y^2 e^x \vec{j}) \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k}) \\ &= x^2 e^y dx + y^2 e^x dy \end{aligned}$$



Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \oint [x^2 e^y dx + y^2 e^x dy]$$

$$= \iint_R [\partial F_2 / \partial x - \partial F_1 / \partial y] dy dx$$

$$= \int_0^2 \int_0^3 (y^2 e^x - x^2 e^y) dy dx$$

$$= \int_0^2 [e^x y^3/3 - x^2 e^y]_0^3 dx$$

$$= \int_0^2 [e^x/3 \times 27 - x^2 e^3 - 0 + x^2] dx$$

$$= \int_0^2 (9e^x - x^2 e^3 + x^2) dx$$

$$= [9e^x - e^3 x^3/3 + x^3/3]_0^2$$

$$= 9e^2 - e^3 \times 8/3 + 8/3 - 9 + 0 = 9(e^2 - 1) - 8/3(e^3 - 1)$$

$$= e^2 - 8e^3/3 - 19/3$$



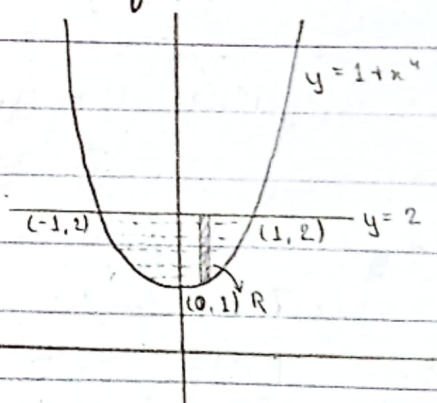
(v) $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (e^y/x, e^y \ln x + 2x)$ and C : region given by $1+x^4 \leq y \leq 2$

Here, region of integration is as shown in figure:

$$\therefore R: -1 \leq x \leq 1, 1+x^4 \leq y \leq 2$$

Also,

$$\begin{aligned} \vec{F} \cdot d\vec{r} &= [e^y/x \vec{i} + (e^y \ln x + 2x) \vec{j}] \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k}) \\ &= e^y/x dx + (e^y \ln x + 2x) dy \end{aligned}$$



Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \oint [e^y/x dx + (e^y \ln x + 2x) dy]$$

$$= \iint_R (\partial F_2/\partial x - \partial F_1/\partial y) dy dx$$

$$= \int_{-1}^1 \int_{1+x^4}^2 (e^y/x + 2 - e^y/x) dy dx$$

$$= 2 \int_{-1}^1 \int_{1-x}^2 dy dx$$

$$= 2 \int_{-1}^1 [y]_{1-x}^2 dx$$

$$= 2 \int_{-1}^1 (2 - 1 - x) dx$$

$$= 2 \int_{-1}^1 (1 - x) dx$$

$$= 2 [x - x^2/2]_{-1}^1$$

$$= 2 [1 - 1/2 + 1 - 1/2]$$

$$= 2 [2 - 1]$$

$$= 2$$



14. Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ using Stoke's theorem.

(i) $\vec{F} = (y^2, 2xy + \sin x, 0)$ and $C: 0 \leq x \leq \pi/2, 0 \leq y \leq 2$.

Here,

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & 2xy + \sin x & 0 \end{vmatrix}$$

$$= (0 - 0)\vec{i} - (0 - 0)\vec{j} + (2y + \cos x - 2y)\vec{k}$$

$$= \cos x \vec{k}$$

The given surface is assumed to be $z = 0$, i.e. the xy -plane. So, normal to this surface is

$$\vec{n} = \vec{k}$$

$$(\nabla \times \vec{F}) \cdot \vec{n} = \cos x$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R \cos x \frac{dx dy}{\vec{n} \cdot \vec{k}}$$

$$= \int_0^2 \int_0^{\pi/2} \cos x \, dx dy$$

$$= \int_0^2 [\sin x]_0^{\pi/2} dy$$

$$= \int_0^2 (1-0) dy$$

$$= [y]_0^2$$

$$= \underline{\underline{2}} \quad \checkmark$$

(ii) $\vec{F} = -5y\vec{i} + 4x\vec{j} + z\vec{k}$ and $C: x^2 + y^2 = 4, z=1$

Here,

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -5y & 4x & z \end{vmatrix} \\ &= (0-0)\vec{i} - (0-0)\vec{j} + (4+5)\vec{k} \\ &= 9\vec{k} \end{aligned}$$

The given surface is $z=1$, i.e. $\phi = z-1$.

$$\therefore \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \vec{k}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = 9$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R 9 \frac{dx \, dy}{\vec{n} \cdot \vec{k}}$$

$$= 9 \int_0^{2\pi} \int_0^2 r \, dr \, d\theta$$

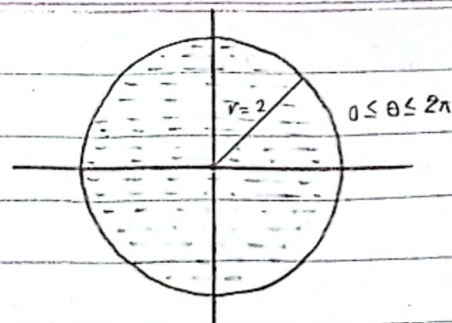
$$= 9 \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^2 d\theta$$

$$= 9 \int_0^{2\pi} (2 - 0) d\theta$$

$$= 18 [\theta]_0^{2\pi}$$

$$= 18 \times 2\pi$$

$$= \underline{\underline{36\pi}}$$



(iii) $\vec{F} = (1, xy, yz)$ and $S: x^2 + y^2 \leq z, y \geq 0, z \leq 4$

Here,

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & xy & yz \end{vmatrix}$$

$$= (z - 0)\vec{i} - (0 - 0)\vec{j} + (y - 0)\vec{k}$$

$$= z\vec{i} + y\vec{k}$$

The given surface is $z = 4$, i.e. $\phi = z - 4$

$$\text{So, } \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \vec{k}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = y$$

Now,

§

(iv) $\vec{F} = (y, \frac{z}{2}, \frac{3y}{2})$ and $C: x^2 + y^2 + z^2 = 6z$, $z = x+3$

Here,

⑩ ④ $\vec{F} = (y, \frac{z}{2}, \frac{3y}{2})$ & $C: x^2 + y^2 + z^2 = 6z$
& $z = x+3$.

Soln

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & \frac{z}{2} & \frac{3y}{2} \end{vmatrix}$$

$$= (\frac{3}{2} - \frac{1}{2})\vec{i} - 0\vec{j} + (0-1)\vec{k}$$

$$= \vec{i} - \vec{k}$$

Put $x=4$, $y=0$, $z=x+3=4+3$.

$\therefore \vec{r} = (4, 0, 4+3)$

$\vec{r}_1 = (1, 0, 1)$

$\vec{r}_2 = (0, 1, 0)$

So, $\vec{N} = \vec{r}_1 \times \vec{r}_2$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= -\vec{i} + \vec{k}$$

So, $(\nabla \times \vec{F}) \cdot \vec{N} = -1-1 = -2$

Here, given surface is,

$x^2 + y^2 + z^2 = 6z$, $z = x+3$

$\Rightarrow x^2 + y^2 + (x+3)^2 = 6(x+3)$

$\Rightarrow x^2 + y^2 + x^2 + 6x + 9 = 6x + 18$

$\Rightarrow 2x^2 + y^2 = 9$

$\Rightarrow \frac{x^2}{9/2} + \frac{y^2}{9} = 1$, which is an eqn of ellipse.

Now

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{N} \, dx \, dy$$

$$= -2 \iint_R dx \, dy$$

$$= -2 \times \text{Area of an ellipse.}$$

$$= -2 \cdot \pi ab$$

$$= -2 \cdot \pi \cdot \frac{3}{\sqrt{2}} \cdot 3$$

$$= -9\sqrt{2} \pi \quad \#$$

(v) $\vec{F} = (y, xz^3, -zy^3)$ and $C: x^2 + y^2 = 4, z = 3$

Here,

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & xz^3 & -zy^3 \end{vmatrix} \\ &= (-3zy^2 - 3xz^2)\vec{i} - (0-0)\vec{j} + (z^3-1)\vec{k} \\ &= (-3zy^2 - 3xz^2)\vec{i} + (z^3-1)\vec{k}\end{aligned}$$

The given surface is $z=3$, i.e. $\phi = z-3$

$$\therefore \vec{n} = \nabla \phi = \vec{k}$$

$$|\nabla \phi|$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = z^3 - 1$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R (z^3 - 1) \frac{dx dy}{\vec{n} \cdot \vec{k}}$$

$$= \iint_R (3^3 - 1) \, dx dy$$

$$= 26 \times \text{area of the circle}$$

$$= 26 \times \pi \times r^2$$

$$= 26 \times \pi \times 2^2$$

$$= 104\pi$$

(vi) $\vec{F} = (2x-y-z)\vec{i} + (x+y+z^2)\vec{j} + (3x-2y+4z)\vec{k}$ and $C: x^2 + y^2 = a^2, z = 0$

Here,

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y-z & x+y+z^2 & 3x-2y+4z \end{vmatrix} \\ &= (-2+2z)\vec{i} - (3+1)\vec{j} + (1+1)\vec{k} \\ &= (-2+2z)\vec{i} - 4\vec{j} + 2\vec{k}\end{aligned}$$

The given surface is $z=0$, i.e. $\phi=z$.

$$\text{No, } \vec{n} = \nabla \phi = \vec{k}$$

$$|\nabla \phi|$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = 2$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R 2 \frac{dx dy}{\vec{n} \cdot \vec{k}}$$

$$= 2 \int_0^{2\pi} \int_0^a r \, dr \, d\theta$$

$$= 2 \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^a d\theta$$

$$= 2 \times \frac{1}{2} \int_0^{2\pi} a^2 d\theta$$

$$= 2a^2 [\theta]_0^{2\pi}$$

$$= 2a^2 \times 2\pi$$

$$= \underline{\underline{4a^2\pi}}$$

(vii) $\vec{F} = (4z, -2x, 2x)$ and $C: x^2 + y^2 = 1, z = y + 1$.

Here,

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4z & -2x & 2x \end{vmatrix} \\ &= 0\vec{i} - (2-4)\vec{j} + (-2-0)\vec{k} \\ &= 2\vec{j} - 2\vec{k}\end{aligned}$$

The given surface is $z = y + 1$, i.e. $\phi = z - y - 1$
 So, $\vec{n} = \nabla \phi = -\vec{j} + \vec{k}$

$$\begin{aligned}|\nabla \phi| &= \sqrt{2} \\ \therefore (\nabla \times \vec{F}) \cdot \vec{n} &= \frac{-2-2}{\sqrt{2}} = \frac{-4}{\sqrt{2}}\end{aligned}$$

Now,

$$\oint \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint -\frac{4}{\sqrt{2}} \frac{dx dy}{\vec{n} \cdot \vec{k}}$$

$$= \iint (-4) \, dx dy$$

$$= -4 \int_0^{2\pi} \int_0^1 r \, dr \, d\theta$$

$$= -4 \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^1 d\theta$$

$$= -4 \int_0^{2\pi} \frac{1}{2} \, d\theta$$

$$= -4 \times \frac{1}{2} [\theta]_0^{2\pi}$$

$$= -2 \times 2\pi$$

$$= \underline{\underline{-4\pi}}$$

(viii) $\vec{F} = (y, z, x)$ and C : the surface of sphere $x^2 + y^2 + z^2 = 1$ above the xy plane.

Here,

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix}$$

$$= -\vec{i} - \vec{j} - \vec{k}$$

The given surface is $x^2 + y^2 + z^2 = 1$, i.e. $\phi = x^2 + y^2 + z^2 - 1$

$$\text{So, } \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2x\vec{i} + 2y\vec{j} + 2z\vec{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}}$$

$$= x\vec{i} + y\vec{j} + z\vec{k} \quad [\because x^2 + y^2 + z^2 = 1]$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = -x - y - z$$

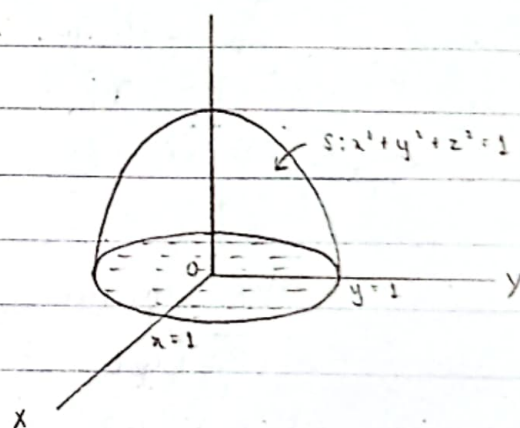
Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R (-x - y - z) \frac{dx \, dy}{\vec{n} \cdot \vec{k}}$$

$$= \iint_R \left(\frac{-x-y}{z} - 1 \right) dx \, dy$$

$$= \iint_R \left(\frac{-x-y}{\sqrt{1-x^2-y^2}} - 1 \right) dy \, dx$$



Let $x = r \cos \theta$, $y = r \sin \theta$, then $dy \, dx = r \, dr \, d\theta$ Then,

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^1 \int_0^{2\pi} \left(\frac{-r \cos \theta + \sin \theta}{\sqrt{1-r^2}} + 1 \right) r \, dr \, d\theta$$

$$= \int_0^1 \int_0^{2\pi} \left(\frac{-r^2 (\cos \theta + \sin \theta)}{\sqrt{1-r^2}} + r \right) d\theta \, dr$$

$$= \int_0^1 \left[-\frac{r^2}{\sqrt{1-r^2}} (\sin \theta - \cos \theta) + r\theta \right]_0^{2\pi} dr$$

$$= \int_0^1 \left(-\frac{r}{\sqrt{1-r^2}} (0-1) + 2\pi r + \frac{r}{\sqrt{1-r^2}} (0-1) - 0 \right) dr$$

$$= \int_0^1 2\pi r dr$$

$$= 2\pi \left[\frac{r^2}{2} \right]_0^1$$

$$= 2\pi \times \frac{1}{2}$$

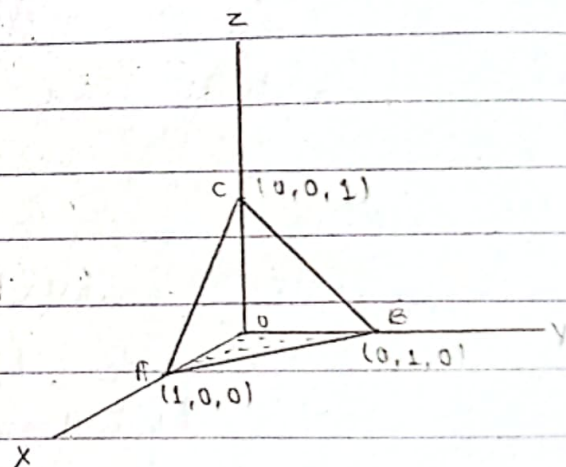
$$= \pi$$

(ix) $\vec{F} = (y^2, 0, x^3)$ and C : boundary of the triangle with vertices $(1, 0, 0), (0, 1, 0), (0, 0, 1)$

✓

Here,

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & 0 & x^3 \end{vmatrix} \\ &= 0\vec{i} - 3x^2\vec{j} - 3y^2\vec{k} \\ &= -3x^2\vec{j} - 3y^2\vec{k} \end{aligned}$$



The equation of plane ABC is

$$x + y + z = 1$$

Hence, the given surface is $x + y + z = 1$, i.e. $\phi = x + y + z - 1$

$$\text{So, } \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = \frac{-3x^2 - 3y^2}{\sqrt{3}}$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} ds$$

$$= \iint_R \frac{1}{\sqrt{3}} (-3x^2 - 3y^2) \frac{dy dx}{\vec{n} \cdot \vec{k}}$$

$$= \int_0^1 \int_0^{1-x} (-3x^2 - 3y^2) dy dx$$

$$= \int_0^1 \left[-3x^2 y - 3 \times \frac{y^3}{3} \right]_0^{1-x} dx$$

$$= \int_0^1 [-3x^2(1-x) - (1-x)^3 - 0] dx$$

$$= \int_0^1 (-3x^2 + 3x^3 - 1 + 3x - 3x^2 + x^3) dx$$

$$= \int_0^1 (4x^3 - 6x^2 + 3x - 1) dx$$

$$= \left[4x^4/4 - 6x^3/3 + 3x^2/2 - x \right]_0^1$$

$$= 1 - 2 + 3/2 - 1 = 0$$

$$= -1/2$$



(x) $\vec{F} = -3y\vec{i} + 3x\vec{j} + z\vec{k}$ and S : along circle $x^2 + y^2 = 4, z = 1$.

$\vec{n}$

Here,

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -3y & 3x & z \end{vmatrix}$$

$$= 0\vec{i} - 0\vec{j} + (3+3)\vec{k}$$

$$= 6\vec{k}$$

The given surface is $z = 1$, i.e. $\phi = z - 1$

$$\text{so, } \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \vec{k}$$

$$|\nabla \phi|$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = 6$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R 6 \frac{dxdy}{\vec{n} \cdot \vec{k}}$$

$$= 6 \int_0^{2\pi} \int_0^2 r \, dr \, d\theta$$

$$= 6 \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^2 d\theta$$

$$= 6 \int_0^{2\pi} 2 \, d\theta$$

$$= 12 [\theta]_0^{2\pi}$$

$$= 12 \times 2\pi$$

$$= \underline{\underline{24\pi}}$$

15. Use Stoke's theorem to evaluate $\oint_C [4ydx + 2zdy + 6ydz]$ where C is the curve of intersection of $x^2 + y^2 + z^2 = 6z$ and $z = x+3$.

✓ Here,

$$\oint_C \vec{F} \cdot d\vec{r} = \int (4ydx + 2zdy + 6ydz)$$

$$= \int [(4y\vec{i} + 2z\vec{j} + 6y\vec{k}) \cdot (\vec{i}dx + \vec{j}dy + \vec{k}dz)]$$

$$\text{i.e. } \vec{F} = 4y\vec{i} + 2z\vec{j} + 6y\vec{k}$$

Then,

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & 2z & 6y \end{vmatrix} \\ &= 4\vec{i} - 4\vec{k} \end{aligned}$$

The given surface is $z = x + 3$, i.e. $\phi = z - x - 3$

$$\text{So, } \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{-\vec{i} + \vec{k}}{\sqrt{2}}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = \frac{-4 - 4}{\sqrt{2}} = -4\sqrt{2}$$

Also, we have,

$$x^2 + y^2 + z^2 - 6z = 0$$

$$\hookrightarrow x^2 + y^2 + z^2 - 2 \cdot z \cdot 3 + 3^2 = 3^2$$

$\hookrightarrow x^2 + y^2 + (z - 3)^2 = 3^2$ is the equation of sphere with centre $(0, 0, 3)$ and radius 3.

Now,

$$\oint \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R -4\sqrt{2} \cdot \frac{dx \, dy}{\sqrt{2} \cdot 2}$$

$$= \iint_R -4\sqrt{2} \times \sqrt{2} \, dy \, dx$$

$$= -8 \times \text{area of circle}$$

$$= -8 \times \pi r^2$$

$$= -8 \times \pi \times (3)^2$$

$$= -72\pi$$

$$= \underline{\underline{72\pi}}$$

16. Find $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (e^{x+y}, e^{x-y})$ and C : boundary of the triangle $x \leq y \leq 2x$, $0 \leq x \leq 1$, by using

(i) Green's theorem

✓ Here, the region of integration is
 $R: 0 \leq x \leq 1, x \leq y \leq 2x$

Also,

$$\begin{aligned}\vec{F} \cdot d\vec{r} &= (e^{x+y} \vec{i} + e^{x-y} \vec{j}) \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k}) \\ &= e^{x+y} dx + e^{x-y} dy\end{aligned}$$

Now,

$$\begin{aligned}\oint_C \vec{F} \cdot d\vec{r} &= \oint [e^{x+y} dx + e^{x-y} dy] \\ &= \iint_R [\partial F_2 / \partial x - \partial F_1 / \partial y] dy dx \\ &= \int_0^1 \int_x^{2x} [e^x e^{-y} - e^x e^y] dy dx \\ &= \int_0^1 [-e^x e^{-y} - e^x e^y]_x^{2x} dx \\ &= \int_0^1 [-e^x e^{-2x} - e^x e^{2x} + e^x e^{-x} + e^x e^x] dx \\ &= \int_0^1 [-e^{-x} - e^{3x} + e^{2x} + 1] dx \\ &= [e^{-x} - e^{3x}/3 + e^{2x}/2 + x]_0^1 \\ &= 1/e - e^3/3 + e^2/2 + 1 - 1 + 1/3 - 1/2 - 0 \\ &= 1/e + e^2/2 - e^3/3 - 1/6\end{aligned}$$

(ii) Stoke's theorem

Here,

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{x+y} & e^{x-y} & 0 \end{vmatrix}$$
$$= (e^{x-y} - e^{x+y}) \vec{k}$$

The given surface is assumed to be $z=0$, i.e. the xy plane. so, normal to this surface is

$$\vec{n} = \vec{k}$$

$$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = e^{x-y} - e^{x+y}$$

Now,

$$\oint \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \iint_R (e^{x-y} - e^{x+y}) \frac{dy dx}{\vec{n} \cdot \vec{k}}$$

$$= \int_0^1 \int_x^{2x} (e^x e^{-y} - e^x e^y) dy dx$$

$$= \int_0^1 [-e^x e^{-y} - e^x e^y]_x^{2x} dx$$

$$= \int_0^1 (-e^{-x} - e^{3x} + e^0 + e^{2x}) dx$$

$$= [e^{-x} - \frac{e^{3x}}{3} + x + \frac{e^{2x}}{2}]_0^1$$

$$= \frac{1}{e} - \frac{e^3}{3} + 1 + \frac{e^2}{2} - 1 + \frac{1}{3} - 0 - \frac{1}{2}$$

$$= \frac{1}{e} + \frac{e^2}{2} - \frac{e^3}{3} - \frac{1}{6}$$

Surface Integral

17. Evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where,

(i) $\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$ and $\vec{r} = (u \cos v, u \sin v, 3v)$; $0 \leq u \leq 1$,
 $0 \leq v \leq 2\pi$

2/

given

$$\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$$

$$\vec{r} = u \cos v \vec{i} + u \sin v \vec{j} + 3v \vec{k}; \quad 0 \leq u \leq 1, \quad 0 \leq v \leq 2\pi$$

$$\vec{r}_u = \cos v \vec{i} + \sin v \vec{j}$$

$$\vec{r}_v = -u \sin v \vec{i} + u \cos v \vec{j} + 3 \vec{k}$$

Then,

$$\vec{N} = \vec{r}_u \times \vec{r}_v$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & 0 \\ -u \sin v & u \cos v & 3 \end{vmatrix}$$

$$= 3 \sin v \vec{i} - 3 \cos v \vec{j} + (u \cos^2 v + u \sin^2 v) \vec{k}$$

$$= 3 \sin v \vec{i} - 3 \cos v \vec{j} + u \vec{k}$$

Also,

$$\vec{F} \cdot \vec{N} = (u^2 \vec{i} + v^2 \vec{j} + w^2 \vec{k}) (3 \sin v \vec{i} - 3 \cos v \vec{j} + u \vec{k})$$
$$= 3u^2 \sin v - 3v^2 \cos v + uw^2$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_R \vec{F} \cdot \vec{N} dA$$

$$= \int_0^1 \int_0^{2\pi} (3u^2 \sin v - 3v^2 \cos v + 9uv^2) dv du$$

$$= \int_0^1 -3u^2 \cos v -$$

9

(ii) $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ and S : the portion of the plane $2x + y + z = 6$ in the first octant.

given
 $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$

Let $x = u, y = v, \Rightarrow z = 6 - 2u - v$. Then, the parametric form of given surface is

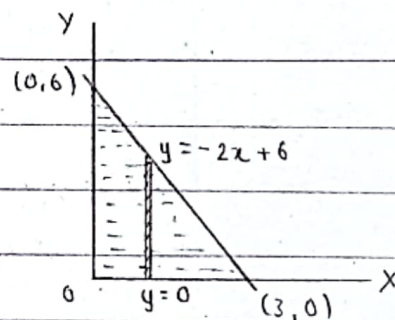
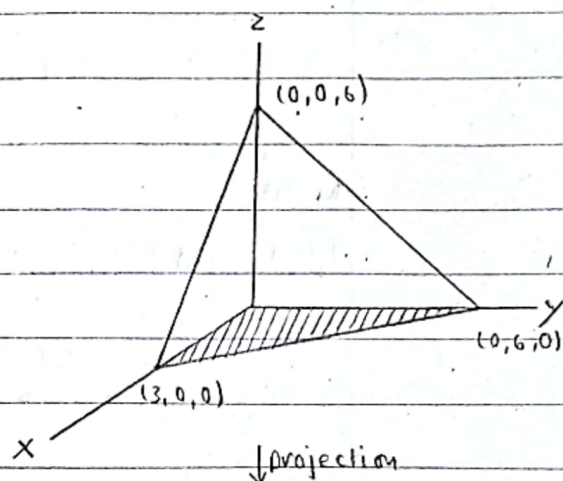
$$\vec{r}(u, v) = (u, v, 6 - 2u - v)$$

$$\therefore \vec{r}_u = (1, 0, -2)$$

$$\vec{r}_v = (0, 1, -1)$$

Then,

$$\begin{aligned} \vec{N} &= \vec{r}_u \times \vec{r}_v \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -2 \\ 0 & 1 & -1 \end{vmatrix} \\ &= 2\vec{i} + \vec{j} + \vec{k} \end{aligned}$$



Also,

$$\begin{aligned} \vec{F} \cdot \vec{N} &= [u\vec{i} + v\vec{j} + (6 - 2u - v)\vec{k}] \cdot (2\vec{i} + \vec{j} + \vec{k}) \\ &= 2u + v + 6 - 2u - v \\ &= 6 \end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iint_R (\vec{F} \cdot \vec{N}) \, dA$$

$$= \int_0^3 \int_0^{2u+6} 6 \, dv \, du$$

$$= 6 \int_0^3 [v]_0^{2u+6} \, du$$

$$= 6 \int_0^3 (2u + 6) \, du = 6 [-u^2 + 6u]_0^3 = 6 \times 9 = \underline{\underline{54}}$$

(iii) $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant.

Here, the given surface is $x^2 + y^2 + z^2 = 1$, i.e. $\phi = x^2 + y^2 + z^2 - 1$

$$\therefore \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2x\vec{i} + 2y\vec{j} + 2z\vec{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\begin{aligned}\therefore \vec{F} \cdot \vec{n} &= (yz\vec{i} + zx\vec{j} + xy\vec{k}) \cdot (x\vec{i} + y\vec{j} + z\vec{k}) \\ &= xyz + xyz + xyz \\ &= 3xyz\end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F} \cdot \vec{n} \frac{dx \, dy}{n_z}$$

$$= \iint_R 3xyz \frac{dx \, dy}{z}$$

$$= 3 \int_0^{\pi/2} \int_0^1 (r \cos \theta \cdot r \sin \theta) r \, dr \, d\theta$$

$$= 3 \int_0^{\pi/2} [\cos \theta \sin \theta \cdot r^4/4]_0^1 \, d\theta$$

$$= 3 \int_0^{\pi/2} \cos \theta \sin \theta \times \frac{1}{4} \, d\theta$$

$$= \frac{3}{4} \times 2 \int_0^{\pi/2} 2 \sin \theta \cos \theta \, d\theta$$

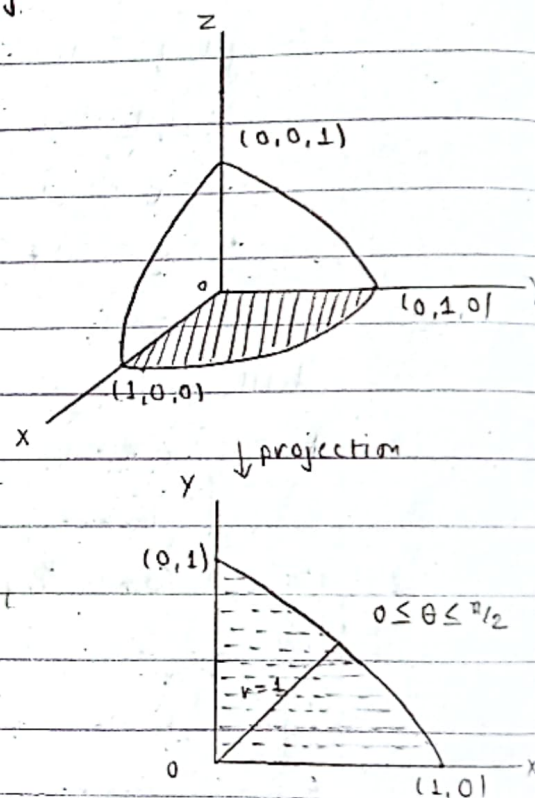
$$= \frac{3}{8} \int_0^{\pi/2} \sin 2\theta \, d\theta$$

$$= \frac{3}{8} \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2}$$

$$= \frac{3}{16} (-\cos \pi + \cos 0)$$

$$= \frac{3}{16} (1 + 1)$$

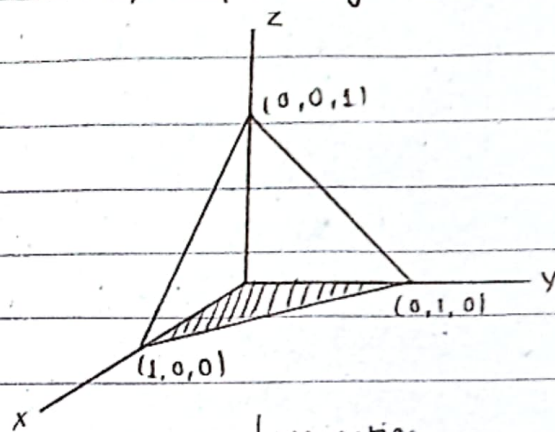
$$= \frac{3}{8}$$



(iv) $\vec{F} = (x^2, 0, 3y^2)$ and S : portion of the plane $x+y+z=1$ in the first octant.

Here, the given surface is $x+y+z=1$, i.e. $\phi = x+y+z-1$
 $\therefore \vec{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$

$$\vec{F} \cdot \vec{n} = \frac{x^2}{\sqrt{3}} + \frac{3y^2}{\sqrt{3}}$$



Now,

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F} \cdot \vec{n} \frac{dy \, dx}{\vec{n} \cdot \vec{k}}$$

$$= \iint_R \frac{1}{\sqrt{3}} (x^2 + 3y^2) \frac{dy \, dx}{1/\sqrt{3}}$$

$$= \int_0^1 \int_0^{1-x} (x^2 + 3y^2) \, dy \, dx$$

$$= \int_0^1 [x^2 y + y^3]_0^{1-x} \, dx$$

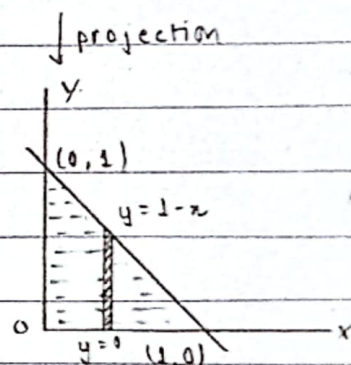
$$= \int_0^1 (x^2 - x^3 + 1 - 3x + 3x^2 - x^3) \, dx$$

$$= \int_0^1 (1 - 3x + 4x^2 - 2x^3) \, dx$$

$$= [x - \frac{3x^2}{2} + 4x^{\frac{3}{3}} - 2x^{\frac{4}{4}}]_0^1$$

$$= 1 - \frac{3}{2} + \frac{4}{3} - \frac{1}{2} = 0$$

$$= \frac{1}{3}$$



(v) $\vec{F} = (3x, 3y, z)$ and S : part of the surface $z=9-x^2-y^2$ with $z \geq 0$.

Here, the given surface is $z=9-x^2-y^2$, i.e. $\phi = z+x^2+y^2-9$

Example 4: Let S be the part of the graph of $z = 9 - x^2 - y^2$ with $z \geq 0$. If $\vec{F} = 3x\vec{i} + 3y\vec{j} + z\vec{k}$. Find the flux of $\vec{F}$ through S .

Solution:

$$\vec{F} = (3x, 3y, z).$$

Given that

And S is part of $z = 9 - x^2 - y^2$ with $z \geq 0$.

Clearly, the projection of the paraboloid in xy -plane is a circle $x^2 + y^2 = 9$.

By surface integral,

$$\iint_S \vec{F} \cdot \vec{n} \, dA = \iint_R (\vec{F} \cdot \vec{N}) \, dx \, dy \quad \dots (i)$$

where,

$$\vec{N} = \vec{r}_x \times \vec{r}_y$$

Since

$$\vec{r} = (x, y, z) = (x, y, 9 - x^2 - y^2)$$

Then

$$\vec{r}_x = (1, 0, -2x), \quad \vec{r}_y = (0, 1, -2y).$$

For the circle, set $x = r \cos \theta$, $y = r \sin \theta$ then $z = 9 - r^2$.

On the circle, radius $r = 3$ and angular variation from $\theta = 0$ to $\theta = 2\pi$.

Here,

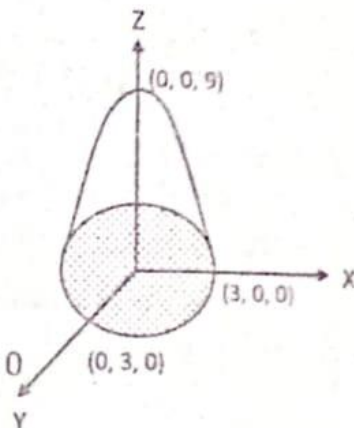
$$\vec{N} = \vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -2x \\ 0 & 1 & -2y \end{vmatrix} = (2x, 2y, 1)$$

Then

$$\begin{aligned} \vec{F} \cdot \vec{N} &= 6x^2 + 6y^2 + z \\ &= 6(x^2 + y^2) + 9 - (x^2 + y^2) \\ &= 6r^2 + 9 - r^2 \\ &= 9 + 5r^2 \end{aligned}$$

Now, (i) becomes,

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, ds &= \int_0^{2\pi} \int_0^3 (9 + 5r^2) r \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{9r^2}{2} + \frac{5}{4} r^4 \right]_0^3 d\theta \end{aligned}$$



$$\begin{aligned} &= \int_0^{2\pi} \left(\frac{81}{2} + \frac{405}{4} \right) d\theta \\ &= \frac{162 + 405}{4} \times 2\pi \\ &= \frac{567\pi}{2} \end{aligned}$$

Thus, $\iint_S \vec{F} \cdot \vec{n} \, ds = \frac{567\pi}{2} .$

18. Find $\iint_S \vec{F} \cdot \vec{n} ds$ using Gauss divergence theorem.

(i) $\vec{F} = y^2 e^z \vec{i} + xy \vec{j} + x \tan^{-1} y \vec{k}$ and $S: x+y+z=1$ in the first octant

Here,

$$\begin{aligned}\nabla \cdot \vec{F} &= (\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z})(y^2 e^z \vec{i} + xy \vec{j} + x \tan^{-1} y \vec{k}) \\ &= 0 + x + 0 \\ &= x\end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x dz dy dx$$

$$= \int_0^1 \int_0^{1-x} x[z]_0^{1-x-y} dy dx$$

$$= \int_0^1 \int_0^{1-x} [x - x^2 - xy] dy dx$$

$$= \int_0^1 [xy - x^2 y - \frac{xy^2}{2}]_0^{1-x} dx$$

$$= \int_0^1 [x - x^2 - x^2 + x^3 - \frac{1}{2}(x - 2x^2 + x^3)] dx$$

$$= \int_0^1 (x - 2x^2 + x^3 - \frac{x}{2} + x^2 - \frac{x^3}{2}) dx$$

$$= \int_0^1 (\frac{x}{2} - x^2 + \frac{x^3}{2}) dx$$

$$= [\frac{x^2}{4} - \frac{x^3}{3} + \frac{x^4}{8}]_0^1$$

$$= \frac{1}{4} - \frac{1}{3} + \frac{1}{8}$$

$$= \frac{1}{24}$$

(ii) $\vec{F} = (4x, x^2y, -x^2z)$ where S is the surface of the tetrahedron with vertices $(0,0,0), (1,0,0), (0,1,0), (0,0,1)$.

Here,

$$\begin{aligned}\nabla \cdot \vec{F} &= 4 + x^2 \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot (4x\vec{i} + x^2y\vec{j} + (-x^2z)\vec{k}) \\ &= 4 + x^2 - x^2 \\ &= 4\end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$

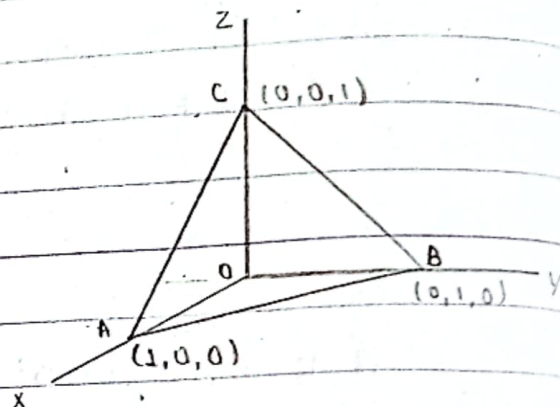
$$= \iiint_V 4 \, dV$$

$$= 4 \times \text{volume of tetrahedron}$$

$$= 4 \times \text{area of } \triangle AOC \times \text{height } OB$$

$$= \frac{4}{3} \times \frac{1}{2} \times 1 \times 1 \times 1$$

$$= \frac{2}{3}$$



(iii) $\vec{F} = (x, 2y, 4z)$ and S : the surface of the cone $\sqrt{x^2+y^2} \leq z$, $0 \leq z \leq 2$

Here,

$$\begin{aligned}\nabla \cdot \vec{F} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (x\vec{i} + 2y\vec{j} + 4z\vec{k}) \\ &= 1 + 2 + 4 \\ &= 7\end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$

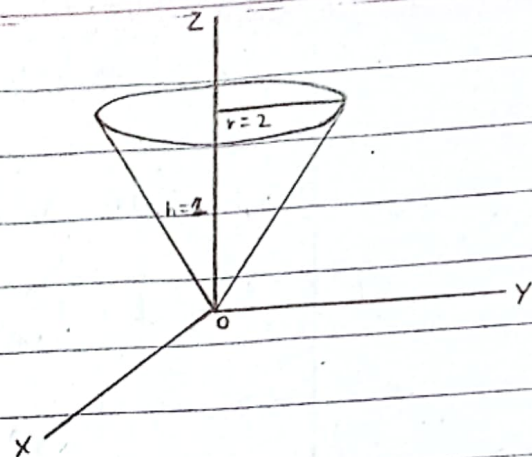
$$= \iiint_V 7 \, dx \, dy \, dz$$

$$= 7 \times \text{volume of cone}$$

$$= 7 \times \frac{1}{3} \pi r^2 h$$

$$= 7 \times \frac{1}{3} \pi \times 4 \times 2$$

$$= \underline{\underline{\frac{56}{3} \pi}}$$



(iv) $\vec{F} = (2x^2, y^2/2, -\cos \pi z)$ and S : tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.

Here,

$$\begin{aligned} \nabla \cdot \vec{F} &= (\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}) (2x^2 \vec{i} + \frac{y^2}{2} \vec{j} - \cos \pi z \vec{k}) \\ &= 4x + y - \cos \pi \end{aligned}$$

Now,

$$\iiint_V \vec{F} \cdot \vec{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$

$$= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (4x+y-\cos \pi) \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (4x+y-\cos \pi) [z]_0^{1-x-y} \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (4x+y-\cos \pi)(1-x-y) \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (4x+y-\cos \pi - 4x^2 - xy + x \cos \pi - 4xy - y^2 + y \cos \pi) \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (4x+y-4x^2-y^2-5xy-\cos \pi + x \cos \pi + y \cos \pi) \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (4x+y-4x^2-y^2-5xy+1-x-y) \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (3x-4x^2-y^2-5xy+1) \, dy \, dx$$

$$= \int_0^1 [3xy - 4x^2y - \frac{y^3}{3} - 5x\frac{y^2}{2} + y]_0^x dx$$

$$= \int_0^1 (3x - 3x^2 - 4x^2 + 4x^3 - \frac{1}{3} + x - x^2 + \frac{x^3}{3} - \frac{5x}{2} + 5x^2 + \frac{5x^3}{2} + 1 - x) dx$$

$$= \int_0^1 (\frac{41}{6}x^3 - 3x^2 + \frac{x}{2} + \frac{x^3}{3}) dx$$

$$= [\frac{41}{6} \times \frac{x^4}{4} - x^3 + \frac{x^2}{4} + \frac{x^4}{12}]_0^1$$

$$= \frac{41}{24} - 1 + \frac{1}{4} + \frac{1}{12}$$

$$= \frac{13}{8}$$

(v) $\vec{F} = (2x^2y, -y^2, 4xz^2)$ and $S: y^2 + z^2 = 3, 0 \leq x \leq 2, y \geq 0, z \geq 0$

Here,

$$\begin{aligned} \nabla \cdot \vec{F} &= (\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}) (2x^2y\vec{i} - y^2\vec{j} + 4xz^2\vec{k}) \\ &= 4xy - 2y + 8xz \end{aligned}$$

Now,

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

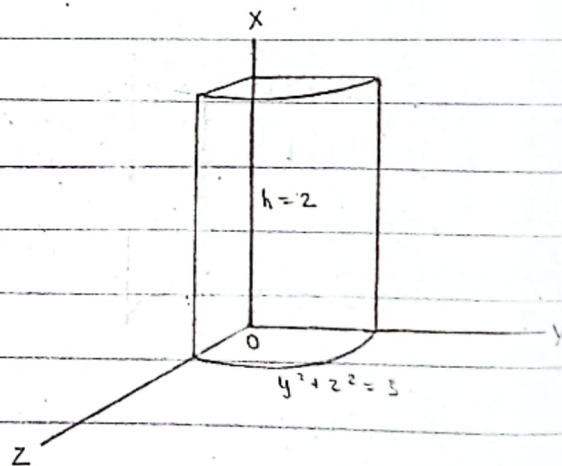
$$= \iiint (4xy - 2y + 8xz) dx dy dz$$

Hints:

$$0 \leq x \leq 2$$

$$0 \leq y \leq \sqrt{3}$$

$$0 \leq z \leq \sqrt{3}$$





19. Evaluate $\iint_S \vec{F} \cdot \vec{n} \, dA$ where $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and S is the surface of the region bounded by $x^2 + y^2 = 4$, $z = 0$, $z = 3$

(i) by using direct method.

Here, the given surface is $x^2 + y^2 = 4$, i.e. $\phi = x^2 + y^2 - 4$

$$\therefore \vec{n} = \nabla \phi = 2x\vec{i} + 2y\vec{j} = x\vec{i} + y\vec{j}$$

$$\therefore \vec{F} \cdot \vec{n} = \frac{|\nabla \phi|}{|\nabla \phi|} = \frac{4x^2 - 2y^3}{\sqrt{4x^2 + 4y^2}} = 2x^2 - y^3$$

Now,

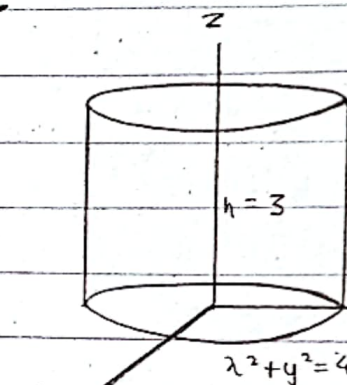
$$\iint_S \vec{F} \cdot \vec{n} \, dA = \iint_R \vec{F} \cdot \vec{n} \frac{dx \, dz}{|\vec{n} \cdot \vec{j}|}$$

$$= \iint_R (2x^2 - y^3) \frac{dx \, dz}{1/2}$$

$$= \iint_R (2x^2 - y^2) \times 2 \, dx \, dz$$

$$= 2 \iint_R (2x^2 - \sqrt{4-x^2}^2 - 4 + x^2) \, dx \, dz$$

$$= 2 \int_0^3 \int_{-2}^2$$



(ii) by using Gauss divergence theorem.

Here,

$$\begin{aligned}\nabla \cdot \vec{F} &= (\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}) (4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}) \\ &= 4 - 4y + 2z\end{aligned}$$

Now,

$$\iint \vec{F} \cdot \vec{n} dA = \iiint_V \nabla \cdot \vec{F} dV$$

$$= \iiint_V (4 - 4y + 2z) dV$$

Let $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, then, $dx dy dz = r dr d\theta dz$. Then

$$\iint \vec{F} \cdot \vec{n} dA = \int_0^3 \int_0^{2\pi} \int_0^2 (4 - 4r \sin \theta + 2z) r dr d\theta dz$$

$$= \int_0^3 \int_0^{2\pi} \int_0^2 (4r - 4r^2 \sin \theta + 2zr) dr d\theta dz$$

$$= \int_0^3 \int_0^{2\pi} \left[2r^2 - \frac{4}{3} r^3 \sin \theta + zr^2 \right]_0^2 d\theta dz$$

$$= \int_0^3 \int_0^{2\pi} \left(8 - \frac{32}{3} \sin \theta + 4z \right) d\theta dz$$

$$= \int_0^3 \left[8\theta + \frac{32}{3} \cos \theta + 4z\theta \right]_0^{2\pi} dz$$

$$= \int_0^3 \left(16\pi + \frac{32}{3} + 8z\pi - \frac{32}{3} \right) dz$$

$$= \int_0^3 (16\pi + 8z\pi) dz$$

$$= \left[16\pi z + 4\pi z^2 \right]_0^3$$

$$= 48\pi + 36\pi$$

$$= \underline{\underline{84\pi}}$$